

Global Business Plan for Drones used in Agriculture – Team Chester

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Phase 1 – Planning the Global Business Enterprise

Identifying Global Business Opportunities (Steven)

There is a multitude of global business opportunities that can be found within Agricultural drones, defining which one best fit the individual pursuit is where it can become a bit difficult and whether it fits within the organization's risk tolerance. For Agricultural drones there are about 4 main routes regarding a global business opportunity; Drones as a service, Input & Service combination, Plantation Operations, and Mapping & Analytics. These four routes of services can bring an array of opportunity to those looking into the field, Drone as a service is simple as it sounds with the rental or leasing of drones to be utilized for agricultural reasoning. The service and usage are outlined to the customer and provide advantages that many other agency techniques have lacked or not provided. Revenue comes from the cost of per acre of coverage the equipment can cover along with the material used and products sprayed (Updates, 2026). Other services can be found to be similar but differ in small ways and or have defining characteristics that do attract as much revenue compared to Drones as a service.

Per-acre spraying costs are driven by type of chemical and application rate, so these factors are inherently variable not fixed. The chemical input itself is the greatest cost driver, since many products of fertilizers, herbicides, insecticides, and fungicides require different amounts and have a range of prices per unit. Chemical costs scale with application rates; this includes specialization or high-concentration chemicals that are applied at higher levels but increase coverage as crops are denser.

Higher spray volumes mean longer flying capabilities, payload cycles, battery swaps and refill times, all of which increases the cost to operate per acre. For example, a heavy nutrient application may involve more effort, energy and equipment use than a low-volume precision application with the same chemical, but its delivery cost per unit might be negligible. Regardless of the chemical type, these operational delivery costs from field setup to software usage, operator labor, and battery depreciation to maintenance are typically a steady but significant per-acre cost (Drones, 2025).

Analyzing International Competitors (Steven)

In the current international market for Agricultural Drones, there is less than a hand full of competitors in the market. There are three main competitors: DJI, XAG Co., and Parrot Drone SAS. These organizations are the top leading in mapping and drone disbursement tech, which are crucial in the current market. DJI also estimates that there will be roughly 400,000 ag drones in operation worldwide by the end of 2024 (DJI Official, 2025). Its greatest strength is the scale and ecosystem of hardware combined with proven software workflows, dealer/service networks, and rapid product iteration all improving the price-performance ratio in a very competitive landscape. Even if operations remain strong elsewhere, its biggest weakness globally is geopolitical and regulatory risk, especially in the US where bans on new DJI imports and models can easily throw sales channels and long-term consumer confidence amok. XAG's strengths include autonomy, swappable task systems (spraying, spreading, mapping and logistics), as well as high-capacity platforms designed for service-provider style operations and fast coverage. These are particularly attractive in large-area, labor-constrained agricultural scenarios. Its issue is that in some geographies, it embodies the same regulatory and cross-border trust issues as Chinese drone companies do, and expanding internationally can only happen by developing local parts support, training and service to take on incumbents.

Parrots European story, in conjunction with years of mapping and data legacy knowledge give them the credibility for businesses that prefer data sovereignty or enterprise procurement needs via precision-ag ops focused on imaging, multispectral collection, and analytics partnerships (Parrot, n.d.). The vulnerability to this strategy is the depth of their spraying platforms and the momentum in each market, Parrot's agricultural presence is more heavily balanced toward mapping/monitoring use cases, as well as older product lines compared to DJI/XAG, which reduces opportunity share in what has been cited as one of the fastest-growing, if not most competitive segments within "spray drone" with operator ecosystems, payload and fill factor being paramount.

Assessing the Economic-Geographic Environment (Steven)

In Africa too, agricultural drones have both systematic constraints and fresh growth to dialectical relationships between an economic-geographical context. Agriculture is still the main sector of employment and a contributor to GDP in many African countries, particularly Sub-Saharan Africa: These women accounted for 90% of agricultural production. But production is often limited due to fragmented landholdings, uneven quality and access to inputs, climate variability and labor constraints due to accelerating urbanization (IA Precision2025).

Geographically, the large acreage and decentralized infrastructure of Africa further augment the capabilities for aerial solutions (Ahmed et al., 2026). The remote geography of many farming regions with poor road connectivity and insufficient logistics network leads to increased cost and time taken to deliver fertilizers, pesticides and advisory services.

Agricultural drones can assist by delivering rapid site-specific input, as well as real-time observation of fields in hard-to-access areas. Drone applications are also stimulated by the climatic variability of desert zones, tropical belts and semi-arid regions; where different circumstances require localized, data-driven interventions rather than standardized treatment techniques.

Agri-tech is increasingly recognized by regional and governmental bodies as a way of enhancing agriculture without replicating expensive Western models. Easier adoption for farmers, reduced operational and maintenance complexity for drone operators, simplified data analytics and more efficient service delivery enable the growth of regional SMEs and service-based business models such as Drone-as-a-Service (DaaS).

Structural inefficiencies, geographically dispersed populations and climate concerns underpin clear demand-side drivers, while the continent's willingness to leapfrog technologies bemuses further scalable deployment models. Despite serious challenges, such as inadequate regulatory capacity and infrastructure in many parts of the continent, the economic and geographic fundamentals render Africa one of the most promising prospects for a burgeoning agricultural drone industry.

Assessing the Social-Cultural Environment (Marilia)

The successful expansion of agricultural drone technology into Africa requires a rigorous understanding of the continent's social and cultural dynamics. Africa is not a monolithic market; its 54 nations encompass more than 3,000 distinct ethnic groups, over 2,000 languages, and profoundly diverse agricultural traditions, spiritual worldviews, and community governance structures (UNESCO Institute for Statistics, 2024). This section assesses the social-cultural environment with particular focus on four primary entry markets: South Africa, Kenya, Nigeria, and Rwanda. It addresses key cultural differences, including the need for

community engagement in rural agricultural contexts and the logistical implications of social acceptance across diverse terrains.

Demographic Trends

Africa's demographic profile constitutes one of the most powerful structural tailwind for agricultural technology adoption globally. The continent's population is projected to reach approximately 1.58 billion by 2026, with sub-Saharan Africa presenting a median age of 19.5 years and over 70% of its population under the age of 30 (Worldometer, 2026). This demographic composition is directly relevant to drone adoption; the 15–29 age cohort, representing an estimated 28–35% of the total population across most sub-Saharan African nations, is characterized by comparatively higher digital literacy, openness to technological innovation, and entrepreneurial orientation in agriculture (International Monetary Fund, 2024).

This youth segment has given rise to a growing class of "agripreneurs", commercially minded farmers who integrate mobile connectivity, digital tools, and market data into traditional farming operations. These represent the most responsive early adopter segment for precision agriculture technology, including drones, and are the primary target for our market entry programs (FAO, n.d).

Literacy rates in key markets further support this opportunity; South Africa registers approximately 94%, Kenya approximately 82%, and both Nigeria and Rwanda approach 70–75% in urban and peri-urban areas, indicating sufficient foundational capacity for technology onboarding when paired with appropriately designed training programs (African Union, 2021).

This demographic advantage is not, however, uniform. Rural agricultural communities particularly those where the average farmer age exceeds 55 years, exhibit considerably greater resistance to technology-mediated farming, rooted in established agrarian routines and skepticism toward external innovation. Effective market penetration requires differentiated

engagement strategies that leverage community authority structures and peer demonstration for older cohorts, while deploying direct digital outreach for younger “agripreneurs”. These two tracks are not mutually exclusive; in many communities, younger “agripreneurs” serve as trusted technology intermediaries for older farmers, creating organic adoption cascades that are both culturally legitimate and scalable (FAO & African Union, 2023).

Cultural Analysis

African cultures are shaped by a distinctive set of values that intersect substantively with how external technologies are evaluated, adopted, and legitimized at the community level. Communalism, the orientation toward collective well-being over individual advancement, is perhaps the most structurally significant cultural factor for drone adoption. Expressed through the Southern African philosophy of ubuntu (“I am because we are”), communalism shapes resource-sharing norms, investment decisions, and social trust in ways that favor cooperative ownership models over individually held capital assets (Phatabantu et al, 2021). Drone programs accessible through cooperative leasing structures align more naturally with this cultural orientation than premium, individually owned equipment.

Language diversity is both a design challenge and an opportunity for competitive differentiation. With over 2,000 distinct languages and English, French, Swahili, Arabic, and Amharic serving as major regional *lingua francas*, drone control interfaces, instructional materials, and customer service infrastructure must be genuinely multilingual to achieve meaningful rural penetration (Eberhard et al., 2023). In East Africa, Swahili-language interfaces dramatically expand accessible user bases. Generic English-only platforms, the current standard for most international drone companies, leave the vast majority of Africa's smallholder farmers unable to operate them independently, creating a direct competitive opportunity for localized solutions.

Gender dynamics are a critical and frequently underestimated dimension of cultural analysis in this context. Women comprise an estimated 60–80% of agricultural labor in sub-Saharan Africa and are responsible for approximately 70% of food production across the continent; yet they own less than 20% of agricultural land and face systematic exclusion from technology adoption programs (FAO, 2023). Drone programs that actively incorporate women as operators, cooperative leaders, and agripreneur trainees simultaneously expand the addressable market and generate social impact outcomes that qualify for development finance instruments, creating commercial advantage.

Spirituality and traditional belief systems are further relevant to technology adoption decisions in rural Africa. In many communities, land holds deep ancestral and spiritual significance; drone surveillance over farmland may be perceived as intrusive or disrespectful. In Nigeria's Yoruba communities, for example, land is ancestral property governed by communal religious protocols (Phatabantu et al., 2021). Cultural sensitivity in drone deployment including community consultation processes, transparent communication of operational purpose, and visible endorsement by trusted local leaders is essential for establishing the social license to operate.

Social Institutions

Social institutions constitute the organizational fabric through which rural African communities make collective decisions, manage shared resources, and evaluate external interventions. The most influential institutions for drone adoption are farmer cooperatives, religious organizations, and community leadership structures (Phatabantu et al., 2021).

Farmer cooperatives, particularly well-developed in Kenya through county-level agricultural associations and dairy cooperatives, provide pre-existing infrastructure for technology co-investment, group purchasing, and shared asset management (U.S. Department

of Agriculture Foreign Agricultura, 2024). These institutions are natural partners for cooperative drone leasing models, offering trust-based distribution channels and built-in accountability mechanisms for equipment stewardship. Rwanda's cooperative-based agricultural extension infrastructure similarly provides a scalable gateway for drone program deployment, as demonstrated by the country's successful integration of drone delivery logistics through cooperative health network partnerships.

Religious institutions command extraordinary social authority and function as de facto community decision-making forums. Engaging religious leaders as early advocates for drone-assisted farming, framing precision agriculture as responsible stewardship of natural resources, can materially accelerate social acceptance in communities where institutional trust in religious leadership exceeds trust in commercial entities. This engagement strategy is not merely symbolic; in Rwanda, church networks played a direct institutional role in post-conflict reconciliation and community development programs, establishing a precedent for faith-institution partnership in development initiatives.

Labor organizations and trade unions, particularly prominent in South Africa's organized agricultural sector, have increasingly engaged with questions of automation and job displacement. Proactive engagement with these institutions, emphasizing drone technology as a complement to, rather than a replacement for, human agricultural labor, and specifically creating drone operator and maintenance technician career pathways, is essential for navigating organized labor dynamics in the South African market and for building broader social credibility across the continent.

Informal Trade Barriers

Beyond formal regulatory frameworks, informal cultural dynamics create substantial market access barriers that require deliberate mitigation. Communication styles in many West

and Central African business contexts are high-context and relationship-oriented: commercial relationships are typically established through extended trust-building processes and community introductions before transactions occur (Phatabantu et al., 2021). Market entry strategies premised on transactional, product-first sales approaches are likely to fail in these contexts. Field representatives require explicit mandates for extended relationship investment, with success metrics weighted toward partnership.

Cultural stigma against foreign technology, particularly technology perceived as surveillance-capable or serving extractive commercial interests, constitutes a recurring barrier. Historical experiences of colonial resource extraction have generated generalized skepticism in many communities toward foreign enterprises that enter without meaningful community partnership or economic reciprocity.

Drone programs that are designed with local equity participation, create local employment, and demonstrate visible reinvestment of economic value within target communities are substantially better positioned to overcome this stigma than programs that present as extractive technology deployments.

E-Commerce Application

Digital market dynamics in Africa provide both a distribution channel and a design benchmark for the drone business. Platforms such as Jumia in Kenya have demonstrated that e-commerce success in African markets requires mobile-first design architecture (given that over 80% of internet access in Africa occurs via smartphone), interfaces adapted to local languages and cultural aesthetics, and pricing mechanisms denominated in local currencies with mobile money integration (Mhlanga & Ndhlovu, 2023).

Summary

Africa's social-cultural environment presents a complex but navigable adoption landscape for agricultural drone technology. Three structural dynamics are most consequential for the proposed business.

First, the continent's youthful, digitally oriented population with over 70% under 30 in sub-Saharan Africa by 2026 generates substantial demand for technology solutions that increase agricultural productivity and create new economic pathways (Worldometer, 2026). Agripreneur certification and training programs, delivered through cooperative networks and mobile platforms, are the most efficient mechanism for converting this demographic potential into market adoption.

Second, communal cultures and cooperative social institutions support collective drone use models but may resist automation perceived as threatening livelihoods or disrespecting traditions. Gender-inclusive program design, proactive labor institution engagement, and community-based governance of drone assets are essential mitigation strategies for these social risks.

Third, informal trade barriers rooted in relationship-first business culture, language diversity, and historical skepticism toward foreign technology require sustained investment in local partnerships, genuine multilingual localization, and community-endorsed introduction strategies.

Table 1
Social-Cultural Factors: Influence on Drone Adoption and Strategic Recommendations

Factor	Influence on Drone Business	Recommendation
Youth Demographics (>70% under 30)	High digital affinity; agripreneur demand for productivity tools	Target agripreneur certification programs in Kenya and Rwanda
Cultural Communalism (ubuntu)	Preference for shared technology and collective ownership	Offer cooperative drone leasing with community governance structures
Language Diversity (2,000+)	Usability barriers with English-only	Develop multilingual apps: Swahili, Hausa,

Factor	Influence on Drone Business	Recommendation
languages)	interfaces	Yoruba, Amharic
Gender Norms (women = 70% of food production)	Systemic exclusion of women from technology adoption programs	Design gender-inclusive pilot programs with women-led demonstrations
Spiritual Land Relationships	Potential perception of drone operations as intrusive	Community consultation and transparent purpose communication
Farmer Cooperatives	Pre-built trust networks; shared asset management capacity	Partner with county cooperatives for distribution and co-investment
Labor Organizations	Job displacement concerns, especially South Africa	Frame drones as complementary; create operator and technician roles
High-Context Communication	Relationship-first business culture; long sales cycles	Invest in local field representatives with 6-month relationship mandates

Note. Synthesis from FAO (2023), Phatabantu et al. (2021), UNESCO Institute for Statistics (2024), and Worldometer (2026).

Module 5: Assessing the Political-Legal Environment

The political and legal landscape of African drone markets is characterized by both significant opportunity and structural complexity. A continent-level assessment must account for Africa's remarkable institutional diversity: 54 sovereign nations with distinct constitutional frameworks, aviation regulatory bodies, trade regimes, data governance systems, and intellectual property enforcement capacities. The drone market projected to reach USD 11.89 billion by 2034 at a 28.10% compound annual growth rate (CAGR) will be shaped as much by regulatory architecture as by technological capability (Market Data Forecast, 2025). This module assesses the political-legal environment across the primary entry markets of Rwanda, Kenya, Nigeria, and South Africa, incorporating initial legal and regulatory considerations identified in earlier work, including the need for harmonized rules on safety and privacy in Africa's diverse regulatory landscape.

Government and Political Environment

Political stability is the foundational prerequisite for sustained technology investment and agricultural development program continuity. Rwanda presents the most compelling

governance environment in Africa for technology-forward business, establishing dedicated drone test corridors, a streamlined online registration portal through the Rwanda Civil Aviation Authority (RCAA), and hosting institutions such as Zipline's Africa headquarters (RCAA, 2020). Rwanda's stable and predictable regulatory posture materially reduces entry barriers and long-term market risk, making it the recommended first-entry market for the proposed drone business.

Kenya offers a similarly favorable investment climate, underpinned by a robust democratic institutional framework and active engagement with the African Continental Free Trade Area (AfCFTA). Kenya's explicit technology and innovation development strategy signals a pro-business, pro-technology governance orientation. The Kenya Civil Aviation Authority (KCAA) has established a structured Remotely Piloted Aircraft Systems (RPAS) regulatory framework, which, while adding compliance steps, signals regulatory maturity and investor protection that is essential for long-term market commitment (KCAA, 2020).

Nigeria represents the largest agricultural market opportunity in Africa, with over 200 million people and an agricultural sector contributing approximately 25% of GDP but introduces substantially higher political risk (Oxford Business Group, 2024). Nigeria scored 25 out of 100 on Transparency International's 2023 Corruption Perceptions Index, placing it in the high-corruption tier (Transparency International, 2024). Drone licensing and operational approval processes in Nigeria's multi-tier regulatory environment, state agricultural agencies, and local government offices, can be subject to discretionary delays and informal cost imposition, requiring significant local compliance expertise and relationship capital to navigate effectively.

South Africa offers the most institutionally sophisticated regulatory environment in sub-Saharan Africa, including an independent judiciary, a well-developed financial services

sector, and a Civil Aviation Authority (SACAA) framework that explicitly governs commercial drone operations under Part 101 of the Civil Aviation Technical Standards (African Development Bank, 2025).

Formal Trade Barriers

Import tariffs on drone technology across African markets average 10–20% of assessed value, representing a material cost component for hardware-centric market entry. Kenya imposes approximately 15% import duty on unmanned aircraft systems with an additional 16% Value Added Tax, bringing the total import tax burden to over 30% on hardware free-on-board (FOB) value (Kenya Revenue Authority, 2024). Nigeria's import duty structure for electronics and precision instruments ranges from 5–20%, with port processing delays and associated informal costs constituting an additional, less predictable cost element.

These tariff barriers present a compelling economic case for local assembly operations in target markets. By establishing local assembly partnerships, even at the level of final assembly and quality testing, the business can access preferential tariff treatment under AfCFTA's rules of origin provisions, potentially reducing effective duty rates to 0–5% for intra-African trade among member states. Rwanda and Kenya have both established Special Economic Zones (SEZs) offering additional tariff and tax incentives for technology assembly operations, further enhancing the financial case for localized manufacturing partnerships (African Union, 2021).

Foreign exchange controls represent a second category of formal trade barrier with direct operational implications. Nigeria's Central Bank has historically imposed strict foreign exchange controls, limiting USD access and creating challenges for profit repatriation. The Nigerian Naira's depreciation of approximately 70% against the U.S. Dollar between 2022 and 2024 has dramatically increased the local-currency cost of imported drone hardware,

compressing buyer affordability (IMF, 2024). Pricing strategies denominated in local currencies supported by cooperative leasing models are essential responses to this structural constraint.

Promoting Global Business and Investment Incentives

African governments have deployed a range of investment incentives designed to attract technology sector foreign direct investment (FDI) that are directly relevant to the proposed drone business. Rwanda offers income tax holidays of up to 15 years for qualifying strategic investments, including agritech firms, under the Rwanda Development Board's (RDB) investment incentive framework (Rwanda Development Board, 2021). Kenya's Special Economic Zones provide corporate income tax rates of 10%, compared to the standard 30% rate, for manufacturing and technology assembly operations, in addition to customs duty exemptions on capital equipment imports.

The African Development Bank (AfDB) authorized approximately USD 11 billion in new investments during 2024, with substantial allocations directed toward agriculture, climate resilience, and digital technology adoption across the continent (African Development Bank, 2025). These multilateral financing flows represent both a source of concessional funding for agritech operations, through grant programs and impact investment vehicles, and an institutional signal of confidence in Africa's agricultural technology sector that underpins the proposed market entry thesis.

AfCFTA, in progressive implementation since 2021, is the most structurally significant trade facilitation development in Africa in recent decades. By eliminating tariffs on approximately 90% of goods among member states and establishing harmonized rules of origin, AfCFTA creates conditions for building a pan-African supply chain and distribution network serviceable from a single regional hub (African Union, 2021). Establishing an

assembly and distribution hub in Rwanda or Kenya provides a scalable gateway to the broader continental market without requiring country-by-country market entry investments.

Intellectual Property Protections

Intellectual property protection for drone technology in Africa is governed by two primary regional frameworks. The African Regional Intellectual Property Organization (ARIPO) covers most Anglophone African countries, while the Organisation Africaine de la Propriété Intellectuelle (OAPI) covers most Francophone African countries. Both frameworks provide patent, trademark, and copyright registration mechanisms that confer protection across their respective member networks through a single application process, significantly reducing the administrative burden of multi-country IP registration (European Commission Africa , 2024).

However, IP enforcement capacity varies dramatically across jurisdictions. South Africa maintains the most sophisticated IP enforcement infrastructure, including dedicated commercial courts and a well-developed trademark registry. Kenya has substantially strengthened its Kenya Industrial Property Institute (KIPI) in recent years. Nigeria, while possessing formal IP registration mechanisms, faces persistent enforcement challenges due to judicial capacity constraints and limited technical expertise in technology IP adjudication (Phatabantu et al, 2021).

The most material IP risks for the proposed drone business are unauthorized hardware replication and software piracy. The proliferation of low-cost Chinese drone component suppliers creates pathways for knockoff hardware that could undermine price positioning. Defensive IP strategies, including patent filing in ARIPO member states for modular hardware innovations, software copyright registration in all target markets, and U.S. export control compliance for sensitive components, are essential baseline investments. Drone software and

copyright protections exist but remain vulnerable to enforcement gaps across African jurisdictions (European Commission Africa IP Helpdesk, 2024).

Rwanda and Kenya represents the most attractive initial expansion geography due to pro-drone regulatory environments, demonstrated government support for agricultural innovation, and established cooperative distribution networks. Nigeria was consistently identified as a high-reward but high-risk secondary market requiring local partnership structures for effective navigation of regulatory complexity and informal trade barriers.

E-Commerce Application

Online drone sales and service platforms in Africa face distinct legal constraints that directly affect revenue model design. In South Africa, the Protection of Personal Information Act (POPIA), which shares core principles with the EU's GDPR, including consent requirements, data minimisation, and accountability obligations, applies to any entity processing personal information within South African territory, meaning drone-captured farm imagery and precision application records collected from South African operations are subject to its compliance framework (Michalsons, 2025). Together, this regulatory requirement increase operational compliance costs but also create a competitive differentiation opportunity: operators who invest in robust data governance can credibly distinguish themselves from competitors whose data practices face growing scrutiny.

Summary

The political-legal environment across Africa's primary drone markets presents a navigable but complex regulatory landscape. Three strategic conclusions emerge from this analysis.

First, stable governments in Rwanda and Kenya offer genuine investment incentives, tax holidays, SEZ benefits, and clear regulatory frameworks, that materially reduce market

entry risk and accelerate operational timelines. AfCFTA provides a multilateral framework for pan-African scaling from a single regional hub, with Rwanda representing the optimal entry point given its regulatory progressiveness, political stability, and demonstrated drone sector development (RCAA, 2020).

Second, tariff barriers averaging 10–20% and foreign exchange controls in key markets such as Nigeria require proactive mitigation through local assembly partnerships and local-currency pricing models. Without these structural adaptations, unit economics in price-sensitive markets will be uncompetitive.

Third, IP protection frameworks through ARIPO exist but require active investment in registration and enforcement monitoring. Local joint ventures by creating legally recognized domestic entities with host-country standing provide both IP enforcement leverage and regulatory navigation advantages that are difficult to replicate through purely foreign entity structures. Investment in IP registration via ARIPO and participation in the African Drone Forum's harmonization advocacy efforts are recommended as complementary strategies for long-term IP protection.

Table 2

Political-Legal Assessment by Primary Entry Market

Dimension	Rwanda	Kenya	Nigeria	South Africa
Political Stability	High	High	Moderate	Moderate (GNU)
Drone Regulation	Progressive (corridors)	Structured (RPAS)	Complex (multi-tier)	Mature (Part 101)
Import Tariff + VAT	~10% + incentives	~31% (duty + VAT)	5–20% + informal	~15% (SEZ available)
IP Enforcement	Moderate (ARIPO)	Moderate (KIPI)	Limited	Strong
Investment Incentives	15-yr tax holiday	10% SEZ corp. tax	Sector grants	Agri-tech incentives
Corruption (CPI 2023)	53/100 – Low	31/100 – Moderate	25/100 – High	41/100 – Moderate

Dimension	Rwanda	Kenya	Nigeria	South Africa
AfCFTA Membership	Full member	Full member	Full member	Full member

Note. CPI scores from Transparency International (2024). Regulatory data from RCAA (2020), KCAA (2020), and SACAA frameworks. CPI scale: higher = less corrupt.

Phase 2 – Organizing for Global Management Information Systems

Selecting a Global Company Structure (Zach)

In constructing a global business plan, establishing guiding objectives is a necessary step in facilitating operational and financial success. Regarding the proposed agricultural drone business, several strategic objectives assist in the development of the business. First, the business will utilize exporting from a centralized hub to enter the agricultural drone market in African market segments. As a result, the lead-time of supply chain distribution from an American or European based producer to consumers in Africa will need to be minimized, while balancing the accompanying costs. The proposed drone business will seek to save around 15-20% by optimizing logistics to decrease lead-time through actions such as increasing buffer stock (New York SBDC, 2025). The business will extract the benefit of responsiveness to shifts in demand, increasing customer satisfaction and increasing accuracy of forecasting. The KPIs for this objective will involve metrics like average days in transit and the frequency of “stock outs” within localized storage facilities in African markets. The second guiding objective pertains to securing regulatory compliance within African markets. The business will strive to promote BVLOS (beyond line of visual sight) authorization which enables drone operators to manage several drones at once and fly them over vast distances. Standard drone flights are only permitted to fly within the visual field of the pilot (Plaza, 2026). A higher ratio of drones to pilots seeks to reduce labor costs. Additionally, flying multiple drones at once promotes scalability within the business, as consumers with larger plots of land can purchase more drones to improve operational efficiency without increasing operational costs. Currently, BVLOS operations within African markets are extremely

limited. Facilitating the prevalence of BVLOS approvals within African markets will assist in providing African consumers with jobs to fly the drones, and flight corridors where the drones will be permitted to fly autonomously will be constructed. Both metrics will function as KPIs to measure operational success. The final primary objective is to partner with at least 50 farm holder cooperatives. Because the base price point of an agricultural drone is \$5,000 and can range up to \$60,000, the financial burden placed upon African farmers will be diminished if an agreement can be reached with cooperatives (Toll Unscrewed Systems, 2022). Cooperatives will purchase a subscription or pay a per-acre fee, allowing farmers to experience the services of the agricultural drones without spending significant capital upfront. This objective seeks to maximize the ability of the business to reach African consumer demand by creating a price-point that reaches many smaller farms while enabling the business to generate profit. Additionally, the trust built within localized markets will hopefully increase demand of whole drone sales, as farmers come to realize that the up-front capital expenditure is justifiable (Katti, 2020). To assess this objective, the business will track hectares serviced each month and subsequent agricultural metrics like percentage reduction in pesticide waste. These three objectives will assist in constructing a viable global business structure that thrives in exporting agricultural drones from a centralized hub to foreign African markets.

Given the strategic objectives above, the actual organizational framework of the business will consist of a structure that combines a centralized department where research and development and manufacturing will take place, while localized divisions within target African segments distribute drones to local consumers. The localized divisions will provide the centralized production center with feedback from consumers, enabling production and manufacturing to cater towards consumer buying criteria.

This global business model of exporting from a centralized hub supports many facets of business operations. Logistically, exporting can minimize capital expenditures by removing the need to construct a production plant in the localized market. Constructing localized production plants is a costly front-end investment that not only takes significant construction time but wraps up significant capital in physical assets. Such construction increases risk for businesses entering the market as the physical assets are not easily liquidated. Another benefit of exporting stems from certain legislation and foreign relations that create favorable conditions for an export business. Regarding agricultural drone implementation, the African Growth and opportunity Act (AGOA) provided many African countries with “duty-free” entry into US markets for some exports (Nedbank, 2022). The agricultural drones fall under the category of Agri-tech and would benefit from such legislation. Export taxes would experience a decrease and continued positive relations serve to benefit such a proposed drone business.

The actual production hub in the developed economies of the United States of the European Union will spearhead many critical functions for an agricultural drone business. For instance, research and development will assess how to produce drones that meet the desires of consumers in localized markets. Drone hardware design and software development will occur here, and the finished products will be shipped to foreign divisions. Because the business will seek to benefit from favorable trade acts like the AGOA mentioned above, the production hub will also monitor the compliance of its drones and manufacturing processes with pertinent international regulations. Moreover, the hub will carry out quality control of finished products to forestall the sale of defective products before shipping costs are absorbed (Ike, 2024).

The regional divisions that serve localized areas in Africa possess different responsibilities than the centralized hub. They primarily function to receive drones from the

centralized manufacturer and producer, and they distribute products to local consumers. Building strong relationships with local entities often constitutes a primary goal for regional operators. Additionally, these regional sites can serve as local technical support centers if consumers experience difficulties with their products or require repairs. Delegating this task of performing repairs to regionalized centers minimizes the need to ship units back to the hub, reducing operational costs and enabling shorter repair times for customers.

The final piece of the global company structure involves the DaaS (Drone as a Service) Success Team. The goal of this branch is to help local consumers understand the capabilities of the product, implement it accordingly, and understand the value added by the products and services provided. On this team, onboarding specialists will assist in training consumers to utilize the product. Typically, this period will last around 90 days or until a predetermined time where value added by the product can be realized (Hull, 2025). Next, a data analyst from the team will extract initial data recorded by the drone and cross-reference it with the local sample sizes to confirm its accuracy. This confirmation will enable a renewal manager to eventually present a consumer with a report at the end of a service contract to highlight the value added by the agricultural drone. The overall functions of the global business structure are further illustrated in the table below.

Operational Phase	Responsible Entity	Key Activity
Design & Mfg	Global HQ (US/EU)	High-precision manufacturing and software encryption.
Export/Import	Trade Compliance Office	Managing "duty-free" entry via AGOA and port logistics.
Distribution	Country Managers	Vetting local agents and managing inventory "carrying costs".
Service Delivery	Local Partners/JVs	Executing the DaaS model (spraying/mapping) for smallholders.

Operational Phase	Responsible Entity	Key Activity
Feedback Loop	GMIS (Global Information System)	Feeding field data (battery wear, crop health) back to HQ R&D.

Financing Sources for Global Business Operations (Jeruson)

The funding of agricultural drone companies is a capital-intensive process, as it involves high technology costs, regulatory requirements, training, and support services. Since agricultural drones are a part of the agriculture industry, which is growing at a fast rate, the sources of funding should be in line with adapting technology, evolving trends and agricultural investment opportunities. We identified three types of fundings which are internal or self-financing, private capital investment and development finance institutions or grants.

1. Internal or Self Financing

For many agricultural technology startups, internal financing will still be the first source of finance. Typically, most of the companies will use internal sources for early-stage activities. Most of the companies choose internal sources of finance as a way to retain high control and business decisions and to gain investor trust.

2. Private Capital Investment

The growth of the agricultural drone market has been very rapid, attracting investment from private investors, primarily focusing on technology segments that are growing rapidly. The market size of the agriculture drone market is expected to reach \$21.59 billion dollars by 2033, growing at a compound annual growth rate of 26.5% during this period, indicating a high profitability potential for investors in this market segment (Grand View Research, 2025). Venture capital investment is considered to be highly suitable for businesses that are using drone as a service, precision spraying, and data analytics.

3. Development Finance Institutions (DFIs)

Development finance institutions highly support agricultural innovation due to its direct impact on food security and sustainability. The African Development Bank authorized around \$11 billion in investments in 2024, with a huge share of funds being allocated to agriculture, technology adoption, and climate change (African Development Bank, 2025). Drones in agriculture match this agenda, as they help minimize pesticide usage, improve irrigation, and enhance crop yields.

Based on the plan of operations of the team, the estimated costs of setting up the business of agricultural drones in one to two African countries would be between \$195,000 and \$315,000 (Drone Business Startup Budget, 2024). For the purpose of this financial plan, the average of estimate cost is \$250,000.

Table 1

Total First-Year Investment Breakdown

Category	Amount
Drone Fleet & Equipment	\$85,000
Applied R&D & Customization	\$20,000
Assemble & Tech Hub Setup	\$30,000
Software & Data Systems	\$15,000
Licensing, Insurance & Compliance	\$35,000
Personnel & Operations	\$50,000
Marketing & Partnerships	\$15,000
Total First-Year Investment	\$250,000

Note. Cost estimates are based on projected first-year international operational expenses for agricultural drone services.

Internal Funding

The Internal funding contributes about 45% of the total capital, which is about \$112,500. Founders investing their own money into the business shows confidence in the business model, which gives more credibility to attract more investors. This type of funding is critical during the initial period of globalization, as the laws, import regulations, and product adoption rates differ from country to country. The higher the level of internal funding, the more the business can be strategic in its operations, which gives more flexibility to the business to cope with the changing environment.

Development Finance Institutions (DFIs) and Grants

Development finance institutions and global institutions have been actively supporting agricultural innovation in emerging markets. The contribution is about 35% of the total capital, which is about \$87,500. The agricultural sector remains a important part of the African continent, supporting a large portion of the African population, and contributing to food security and development. The African Development Bank approved around \$11 billion worth of investment projects approved for the 2024-2025 period, with a large portion of the investment dedicated to the agricultural, climate, and technology sectors (African Development Bank, 2025). In another development, the World Bank has agreed to increase agricultural finance to \$9 billion a year by 2030 (Reuters, 2024). Drone technology in the agricultural sector fits well into these development priorities, as it has the potential to improve crop yields, reduce the use of pesticides, improve water efficiency, and promote climate-resilient agricultural practices. The strategic importance of grant funding is its ability to decrease the overall risk of starting up a new business, as there is no payback or equity consideration, which may act as a burden in the early stages of the business when the revenue stream is not stable. The risk level is not high because grants do not require interest payments or profit sharing, they reduce the company's financial pressure. This lowers the overall cost

of capital for the business and improves cash flow. As a result, the company can focus on growth and operations instead of worrying about paying back investors or lenders.

There are a number of agricultural technology companies that have recorded successes in the use of grants as a way to finance the expansion of their businesses. Precision Development has received grant financing from the Gates Foundation to finance the expansion of its digital agricultural advisory services (Gates Foundation, 2023). Another example of an agricultural technology company, Hello Tractor, has used development assistance funding to scale up its agricultural mechanization services in Africa (MIT Solve, n.d.).

Private Capital Investment

About 20% of the total amount, which is equal to \$50,000, is obtained through private capital. The global agricultural drone market is expected to grow at a compound annual growth rate (CAGR) of 26.5% by the year 2033. Therefore, the agricultural drone industry has significant profitability potential (Grand View Research, 2025). Private capital investment is important not only due to the capital received but also due to the strategic benefits such investments provide to the enterprise. Private capital provides the enterprise with the ability to grow and also the expertise to grow the enterprise to the global level. Aerobotics is a precision agriculture company based in South Africa. The company has received venture capital to expand the services provided by the enterprise to the global level to include the analysis of crops using drones (Crunchbase, 2025). Therefore, the agricultural drone industry is viewed favorably by venture capitalists, implying the significant profitability potential of the industry.

Creating a Global Management Information System (Zach)

To overcome the "information asymmetry" that often occurs within a global hub and localized distributor model, the firm will implement a centralized Global Management Information System (GMIS). This system will operate as a "Single Source of Truth" (SSOT), ensuring that production standards in the US or the EU are synchronized with real-time market shifts in the localized African segments. The GMIS assists in balancing the initiative to decrease lead-time without decreasing product quality by providing the Western hub with increased visibility into the operational health of the proposed drone business. The GMIS will consist of several critical layers, each addressing a potential bottleneck in the global structure.

One layer is comprised of asset management and predictive maintenance. This layer monitors the real-time data of every drone in a fleet. The data measured includes battery cycles, motor vibrations, and sensor calibration, and the system will execute predictive maintenance algorithms in order to identify potential failures before they occur (Hosdurg, 2026). This triggers the shipment of "modular repair kits" from the US hub to the local technician weeks before a breakdown, aiding to decrease the export lead time and maintain a high utilization rate of the drone fleet.

Another layer involves establishing the Drone-as-a-Service model which is key to helping consumers realize the value of the drones. Essentially, the GMIS will process raw data from drone flights and produce "Prescription Maps" that will be utilized to create value realization reports (Neil, 2025). A member from the aforementioned DaaS success team will then present the report to the consumer and communicate the returns of the investment into the agricultural drone usage. A secondary goal after communicating value realization is to convert the consumer, presumably a cooperative, to transition into a whole purchase consumer. Additionally, the GMIS will provide a customer health score for each farm cooperative customer, alerting the DaaS Success Team to intervene when necessary. This

initiative further assists cooperative consumers in climbing the value chain and eventually owning a product individually.

Overall, the layers within the GMIS strengthen the global supply chain, promoting financial performance and preserving customer satisfaction.

Identifying Human Resources for Global Business Activities (Laura)

Howard V. Perlmutter's introduced in his article, "The Tortuous Evolution of Multinational Corporation" in 1969, a model that establishes an applicable framework for deciding the type of employees for the international business operation in Africa. This framework suggests three and later revised to be four different types methods a company can pursue in deciding personnel to hire. The model includes ethnocentric, polycentric, geocentric and regiocentric. Ethnocentric is companies that are home-country oriented, in this case our business would be USA hiring oriented. Polycentric would-be host-country oriented, so hiring personnel within Africa. Geocentric would be hiring from any country, looking for any individual that could accomplish the job well (Stopford & Wells, 1972, p. 298-302). The last is regiocentric, which is selecting specific regions for operations with countries that have similar cultures and structures (Zeeman, 2025).

All this to say, depending on if the company chooses greenfield investment or acquisition will affect the type of employees hired. If our company chooses to acquire Terra Industries, Aerobotics, PACSys or Flygrow Africa, then employees will be local, also known as polycentric, and ultimately pre-trained in skills needed for execution of operations. If our company chooses greenfield investment, a hybrid of polycentric and geocentric would allow our business to gain trust from the local market and create strategic, advanced designs by hiring talent globally for the drones.

The employees needed for this operation can be evaluated by looking at competitors Aerobotics. While Aerobotics does not publicly disclose the employees involved with

business operations, some information can be seen on “Zoominfo” which shares the business has 74 employees involved in their operation. Below is a chart that states the department, top needed positions based on Aerobotics partially disclosed information (Zoominfo, n.d.). Additionally, under the “Requirement” tab, it states if our company will likely recruit the department in country or out of country and why.

Department	Employee Title	Recruitment
Engineering & Technical	Chief Technology Officer Head of AI Data and Operations Head of Engineering Product Manager	Geocentric for the specialized skills
Sales	Head of Customer Success Business Development, Manager Customer Success Business Development Manager	Polycentric for the country buy-in
Operations	Chief Operating Officer General Manager – Regional General Manager – Actions and Insights Processing Operations Head	Polycentric to operate within country norms and culture
Marketing	Head of Marketing Chief Commercial Officer Chief Growth Officer	Polycentric for specialized campaigns
C-Suite	Chief Executive Officer Chief Operating Officer Chief Technology Officer GM Africa	Ethnocentric as USA investors.
Finance	Group Manager, Accounting Senior Financial Admin	Ethnocentric and Polycentric to maintain tariffs and currency exchanges
Human Resources	HR Manager (Not stated on ZoomInfo)	Geocentric for cost-effective labor

Managing International Financial and Business Risks (Marilia)

International expansion into Africa's agricultural drone markets carries a distinctive risk portfolio shaped by macroeconomic volatility, institutional fragility, social resistance

dynamics, and geopolitical complexity. A rigorous risk management framework must go beyond risk enumeration to include systematic probability and impact assessment, and the identification of operationally viable mitigation strategies grounded in evidence. This section presents a risk assessment across economic, social-cultural, and political-legal dimensions, integrating infrastructure and logistics risks identified in prior work, and develops a structured management response framework appropriate to Africa's market context.

Economic and Financial Risks

Currency risk represents the most immediate and quantifiable financial exposure in the proposed Africa market entry. The Nigerian Naira depreciated approximately 70% against the U.S. Dollar between 2022 and 2024, and the Kenyan Shilling experienced approximately 25% depreciation over the same period (IMF, 2024). For a U.S.-manufactured product priced in USD, these depreciations translate directly into equivalent local-currency price increases, threatening to place drone hardware beyond the reach of smallholder buyers who earn in local currencies. Mitigation strategies include: (a) local-currency revenue models with cooperative leasing priced; (b) natural hedging through local cost base development, including locally employed field technicians and locally sourced operational services; and (c) forward currency contracts for USD-denominated revenue repatriation on contracts exceeding threshold values.

Infrastructure deficits represent a second major economic risk category, with direct implications for the drone's operational value proposition. Sub-Saharan Africa's logistics infrastructure gap, characterized by poor rural road networks, intermittent electricity, limited connectivity, and unreliable cold chain capacity, imposes both direct operational costs and indirect adoption barriers (World Bank Infrastructure, 2020). A drone platform that cannot be serviced due to parts unavailability or technician inaccessibility delivers zero value during a farming season, generating reputational damage. The proposed offline-capable, modular

hardware design is a direct product-level response to this infrastructure risk, reducing dependency on connectivity and enabling community-level maintenance.

Inflation risk is particularly pronounced in Nigeria, where the average Consumer Price Index (CPI) inflation rate reached approximately 28% in 2024, and Zimbabwe, where triple-digit inflation persists (IMF, 2024). In local-currency-priced contracts, inflation erodes real revenue and compresses margins on multi-year service agreements. Price escalation clauses indexed to local CPI, incorporated into cooperative leasing contracts from inception, provide contractual protection against inflation erosion without requiring contract renegotiation.

The Africa agricultural drone market was valued at approximately USD 1.28 billion in 2025 and is projected to reach USD 11.89 billion by 2034, representing a 28.10% CAGR (Market Data Forecast, 2025). While this growth trajectory underscores the opportunity, it also signals a competitive intensity risk: rapid market growth will attract well-capitalized incumbents, including DJI Agriculture and XAG, who may respond to competitive pressure with aggressive pricing or increased Africa-specific product adaptation. Early market penetration and cooperative partnership lock-in are essential for establishing defensible market position before intensified competitive entry.

Social and Cultural Risks

Technology displacement anxiety, the fear that drone automation will eliminate agricultural employment, represents a significant social risk with the potential to generate organized community resistance, negative media coverage, and reputational damage. Youth unemployment in sub-Saharan Africa averages approximately 30%, with rates exceeding 60% in some urban peripheries, making technology-driven displacement a politically and socially sensitive issue (International Labour Organization, 2024). The risk is not purely perceptual: a single agricultural drone can perform the daily spraying equivalent of 30–50 manual laborers.

The mitigation strategy must proactively reframe drones as employment creators in drone operator, maintenance technician, and agricultural data analyst roles, while designing workforce transition programs for communities where displacement effects are material.

Privacy and surveillance concerns represent a growing social risk as African communities become more aware of drone data collection capabilities. Transparency deficits, rather than opposition to the technology itself, are the primary driver of community mistrust, and deployment without agreed data governance rules can escalate into accusations of surveillance and spying. Agricultural data privacy concerns further compound this, as farmer reluctance to share precision data with agribusiness platforms constitutes a documented barrier to digital adoption across the sector (Mhlanga & Ndhlovu, 2023). Community communication programs that demonstrate data use policies, establish data ownership provisions, and restrict non-agricultural data collection are therefore essential trust-building investments.

Gender exclusion risk, if unmanaged, simultaneously limits market penetration and undermines eligibility for development finance instruments. Given that women produce approximately 70% of Africa's food but access less than 20% of formal agricultural services, a drone program that fails to integrate women as operators and beneficiaries will both miss the majority of its target market and underperform on the social impact metrics required for AfDB, USAID, and impact investment qualifying criteria (FAO, 2023).

Political and Legal Risks

Regulatory inconsistency, the risk that drone regulations will change unpredictably, be inconsistently enforced, or be captured by incumbent interests, is a material risk in markets like Nigeria, where approval processes span multiple regulatory tiers. Mitigation requires early regulatory engagement prior to commercial launch, participation in industry associations such as the African Drone and Data Academy (ADDA) network, and modular hardware and

software architecture enabling rapid regulatory adaptation without requiring complete product redesign.

Corruption risk in licensing, customs clearance, and contract enforcement poses financial, reputational, and legal risks including potential exposure under the U.S. Foreign Corrupt Practices Act (FCPA) for U.S.-incorporated entities operating in high-corruption markets. Nigeria's CPI score of 25/100 and several other target markets below the 40/100 threshold are indicators of elevated exposure (Transparency International, 2024). The anti-corruption protocol must include: comprehensive FCPA compliance training for all field personnel; engagement of independent local legal counsel for licensing and approval processes; transparent documentation of all government interactions; and partner selection criteria that include demonstrated integrity records.

Political instability risk, including governance disruptions, election-cycle policy volatility, and the risk of host government policy shifts restricting U.S. technology imports in favor of Chinese alternatives, is a lower-probability but high-impact risk. AfCFTA's multilateral framework and the African Union's explicit preference for technological sovereignty and diverse supply chain partnerships provide partial structural protection against bilateral geopolitical disruption. The proposed modular hardware design, enabling local assembly and component substitution, provides additional product-level resilience against technology import restrictions.

Risk Management Techniques

The proposed integrated risk management framework employs five primary mitigation mechanisms, selected for operational feasibility in emerging market contexts and evidence-based effectiveness in comparable technology deployment programs.

First, public-private partnerships provide blended finance structures that share market entry risk between the company, host governments, and development finance institutions. The World Bank Group has committed to doubling its agricultural finance commitments to USD 9 billion annually by 2030, targeting technologies that improve productivity and climate resilience (Reuters, 2024). Co-investment structures with the AfDB, which authorized USD 11 billion in 2024 investments, can provide concessional financing, first-loss capital tranches, and political risk insurance that materially reduce financial exposure in high-risk markets (African Development Bank, 2025).

Second, local partnership and joint venture structures provide both political risk mitigation and market access acceleration. Local equity partners with established government relationships reduce regulatory friction, provide social capital for community engagement, and create legally recognized domestic entities with host-country standing that may access preferential treatment in public agricultural procurement. In Nigeria, where local equity requirements are most prevalent, minority local equity structures, while maintaining management control through contractual mechanisms, represent the optimal risk-access balance.

Third, insurance mechanisms covering political risk, cyber threats, climate events, and equipment loss provide financial backstops for specific insurable risk categories. Political risk insurance from the Multilateral Investment Guarantee Agency covers expropriation, currency inconvertibility, and political violence—the three highest-severity political risk scenarios. Cyber insurance for drone platform data breaches is increasingly essential as drone-collected farm data achieves commercial value and regulatory scrutiny, particularly under South Africa's POPIA framework.

Fourth, portfolio diversification across the four target markets, Rwanda, Kenya, Nigeria, and South Africa, reduces single-market concentration risk. Each market's risk profile is sufficiently distinct that adverse events in one market are unlikely to be simultaneously replicated across all four. Rwanda's political stability and regulatory progressiveness provide a reliable operational anchor; Nigeria's scale justifies higher risk tolerance; South Africa's institutional sophistication supports premium-segment development; Kenya's cooperative infrastructure provides the best model for mass-market scalability.

Fifth, adaptive product design, including offline capability, modular hardware, multilingual interfaces, and cooperative-compatible ownership models, constitutes a product-level risk mitigation strategy that reduces the impact of infrastructure deficits, cultural barriers, and regulatory variability on operational performance. This architectural philosophy, applied from product inception rather than as a post-market-entry retrofit, is the single most cost-effective risk mitigation investment available to the proposed business.

Supplementary Activity

An interview conducted with an African drone operator with field experience across East and West Africa identified political instability and unpredictable regulatory enforcement as the most operationally disruptive risk categories, consistent with the analysis above. The operator specifically cited local alliance structures, established before market entry rather than in response to problems, as the most effective single risk management mechanism, noting that local partners with regulatory relationships were able to resolve compliance issues in days that would otherwise have delayed operations for months.

E-Commerce Application

Digital sales and service platforms for the drone business require dedicated cybersecurity and data governance risk management frameworks. E-commerce sites handling

drone configuration data, farm mapping records, and customer financial information represent high-value targets for cyberattacks, particularly in markets with limited cybersecurity infrastructure. Encrypted platform architecture, multi-factor authentication for all user accounts, and cybersecurity insurance covering data breach liability are minimum standards for responsible digital operations in African markets. These investments also generate direct commercial value by differentiating the platform's data trustworthiness from competitors—particularly Chinese platforms subject to the National Intelligence Law in markets increasingly sensitive to data sovereignty concerns.

Summary

The risk landscape for agricultural drone operations in Africa is substantial but manageable through systematic, evidence-based risk management frameworks. Three categories demand priority attention.

First, economic risks, led by currency depreciation in Nigeria and Kenya, infrastructure logistics gaps, and inflation erosion, require financial engineering solutions (local-currency pricing, CPI escalation clauses, hedging) combined with product design responses (offline capability, modular maintenance) that reduce dependency on Africa's infrastructure weaknesses.

Second, social risks, including job displacement anxiety, privacy concerns, and gender exclusion, require proactive community engagement programs, gender-inclusive design, and workforce transition investment as non-negotiable market entry prerequisites rather than optional social responsibility additions.

Third, political and legal risks including regulatory inconsistency, corruption exposure, and geopolitical volatility are most effectively managed through the PPP model, local joint

venture structures, and multilateral institutional partnerships that provide both financial protection and political legitimacy in host-country markets.

Table 3

Risk Register: Agricultural Drone Operations in Africa

Risk Category	Specific Risk	Probability	Impact	Primary Mitigation	Evidence Base
Economic	Currency depreciation (Naira, KES)	High	High	Local-currency leasing; CPI-indexed contracts; FX hedging	IMF (2024) – Naira -70% (2022–2024)
Economic	Infrastructure logistics deficits	High	Moderate	Offline-capable hardware; modular repair design; regional service hubs	World Bank Infrastructure (2020)
Economic	Inflation erosion of local margins	Moderate	Moderate	Price escalation clauses indexed to local CPI	IMF (2024) – Nigeria CPI 28%
Economic	Intensified competitive entry	Moderate	Moderate	Early cooperative lock-in; first-mover local partnerships	Grand View Research (2025)
Social	Job displacement resistance	Moderate	High	Drone operator/technician training; employment reframing	ILO (2024) – 30% youth unemployment
Social	Privacy and surveillance concerns	Moderate	Moderate	Offline data processing; community data ownership policies	Phatabantu et al. (2021)
Social	Gender exclusion limiting adoption	Moderate	Moderate	Gender-inclusive pilot programs; women-led cooperative demos	FAO (2023)
Political	Regulatory inconsistency	High	Moderate	ADDA membership; modular compliance architecture	RCAA (2020); KCAA (2020)
Political	Corruption / FCPA exposure	Moderate (NGA)	High	FCPA training; independent legal counsel; partner vetting	Transparency International (2024)
Political	Geopolitical trade disruption	Low	High	Local assembly; AfCFTA reliance;	African Union (2021)

Note. Probability and impact ratings are qualitative assessments based on referenced literature and market intelligence. NGA = Nigeria-specific risk concentration. *Nigeria concentration reflects CPI score of 25/100 (Transparency International, 2024). Table generated with the assistance of Claude (Anthropic, 2026).

Phase 3 – Implementing the Global Market Plan

Product and Target Market Planning for Foreign Markets (Justin)

For an international agricultural drone service targeting smallholder farmers in African markets, the proposed product can be defined as a “drone-as-a-service” offering that combines spraying, mapping, and advisory support rather than selling hardware alone. This service can include precision spraying of pesticides and fertilizers, high-resolution crop scouting, and geospatial mapping that identifies nitrogen deficiencies, weed pressure, and water-stress zones, allowing farmers to apply inputs more selectively (IA Precision, 2025; MarketsandMarkets, 2025). Field evidence from pilots in Kenya, Nigeria, and Malawi suggests that such drones have been associated with reductions in pesticide use, shorter spraying times, and measurable yield gains—for example, one Kenyan maize cooperative using sprayer drones reported about a 50% reduction in pesticide use and a 25% increase in yields over two seasons, while a Nigerian rice project reported over 30% yield improvements after optimizing fertilizer application (IA Precision, 2025; Rural Hack, 2025). The primary customer benefits therefore include potential cost savings on inputs, time savings during spraying operations, reduced exposure of workers to chemicals, and improved crop performance through more accurate and timely interventions (IA Precision, 2025; Farmers Weekly, 2025).

The target market can be described as small to medium-scale commercial farmers and organized smallholder cooperatives in countries where drone regulations and agtech ecosystems are emerging, such as South Africa, Kenya, and Nigeria (GreenCape, 2025; Market Data Forecast, 2025). In South Africa, for instance, agricultural drone revenues were estimated at roughly 1 billion rand in 2024 with projections to more than triple by 2030, indicating growing demand for field monitoring, precision spraying, and data analytics rather than only hardware purchases (GreenCape, 2025). Similar demand trends are suggested at the

global level; industry reports project the agriculture drones market to reach about USD 10.76 billion by 2030, with notable growth of “drone-as-a-service” models that lower upfront capital requirements for farmers (MarketsandMarkets, 2025; Allied Market Research, 2025). For smallholders who often face credit constraints, a pay-per-hectare service model, supported by cooperatives or agribusiness partners, can make access to drone technology more feasible than direct ownership and aligns with observed initiatives where agtech startups bundle drone services with input finance and advisory support (IA Precision, 2025; Market Data Forecast, 2025). In this context, the key product attributes—precision application, bundled analytics, and local operator training—are positioned to address specific pain points related to labor bottlenecks, high input prices, and climate-related production risks in the target markets (GreenCape, 2025; IFPRI, 2025).

Designing a Global Distribution Strategy (Laura)

To execute this global business operation, effective distribution channels and wise intermediaries is essential to the company’s health and success. To better understand this, looking at competitor acquisition opportunities, gives context on the different intermediaries needed within each.

Company Acquisition Opportunities

Company	Location	Year Began	Valuation	Notes	Source
Terra Industries (Terrahaptix)	Nigeria	2023	\$11.75M USD raised in funding.	Defense technology, Agricultural drones, and surveillance. Large local African connections, local governments. (Easy entry)	(TechCebal, 2026) (TerraIndustries, 2026)
Aerobotics	South Africa	2014	\$17M in funding raised	Agricultural related drones.	(Crunchbase, 2025)
PACSys	Ballito, South	2017	No public data	Agricultural Drone Specialists	(PACSys, n.d.)

	Africa		posted		
Flygrow Africa	Kenya	No Date Listed	Startup, no pricing published anywhere yet.	200+ acres sprayed, 32+ farmers registered, 3,000 soil tested.	(Flygrow Africa, n.d.)

Aerobotics, for example partners with local drone pilots to offer services locally in Africa. (Aerobotics, n.d.). Additionally, the company utilized Advexure, which is a large drone reseller. This allows Aerobotics to utilize local labor and sell drones locally without having any inventory on hand (PR Newswire, 2023). Aerobotics operates more in providing data analytics to give farmers the capability to make informed decisions. This company utilizes a third party approach by utilizing intermediaries, such as Advexure, in order to complete their operations.

Whereas competitor like Flygrow Africa is a full in-person service that requires local attention to farms to conduct drone flying, spraying, fertilization and crop monitoring (Flygrow Africa, n.d.). This business operates as through a direct service distribution model, not maximizing their intermediary opportunities. Investing in Flygrow as an acquisition is an opportunity for our company to gain the emerging startup as an intermediary for our company, allowing there to be local distribution within Africa.

In either business model, both distribution channels and intermediaries are necessary for the business operation, displayed in different avenues. Regardless, for both businesses, its dependent upon local employees to manage the drones.

Planning a Global Promotion Strategy (Jeruson)

The development of an effective promotional strategy for the entire globe is important for introducing agricultural drone services in the markets of Africa. The agricultural drone market globally is expected to grow up to a value of 21.59 billion dollars by 2033, with a

compound annual growth rate (CAGR) of 26.5% (Grand View Research, 2025). The agricultural drone market in Africa has a value of 1.28 billion dollars and is expected to grow at a CAGR of 28.1% during the upcoming decade (Market Data Forecast, 2026). Despite the growth of the market, its adoption is dependent upon awareness, economic justification, and trust. Therefore, the promotional strategy for the entire globe will be based upon a combination of digital media marketing, cooperative promotion, and engaging with institutions and government officials.

Digital Media Marketing

Digital marketing is proposed as an essential promotional tool given the rise of mobile and internet penetration in Sub-Saharan Africa. Agricultural technology companies like Aerobotics use digital marketing as a tool for managing client relationships, providing service-related information, and processing client inquiries (Aerobotics, n.d.).

An allocation of around \$6,000 annually is proposed for digital marketing. The digital ads would emphasize the economic benefits associated with the service, including reduced costs and improved productivity. It is worth noting that agricultural drone spraying has the potential to reduce chemical use by as much as 30%, which means farmers can save costs during the season (Avary Drone, 2025). The ads would highlight the economic benefits of the service, such as “Reduce chemical costs by up to 30% per season.”

If the firm can receive 400 online inquiries annually and assuming a conversion rate of 5%, digital marketing can lead to the signing of 20 service contracts annually. This would mean the firm can earn around \$180,000 annually, given the average contract value is \$9,000 per farm. This is an indication of the potential financial benefits of digital marketing compared to the allocated marketing budget.

Partnership and Cooperative Promotion

The second strategic approach is the need for collaboration with agricultural cooperatives, agribusiness suppliers, and local farming associations. It is estimated that about 70% of the total population of Africa depends on the agricultural sector for livelihood (Ayim et al., 2020). The farmers operate in groups, and decisions regarding the purchase of goods are made collectively. This approach is beneficial as the costs of acquiring customers are reduced. An allocation of about \$5,250 annually is proposed, This amount is meant for the promotion through cooperatives. The approach is beneficial as it takes into account the trust relationships developed through the cooperative approach. The rate of adoption is high, especially in rural areas, as the process of adopting new technology is slow (Ayim et al., 2020).

Promotion through cooperatives with an average membership of 50-100 farmers would lead to the conversion of about 20% of the total farmers in one cooperative. This would lead to the conversion of 10-20 farms, which would earn the company an amount ranging from \$90,000 to \$180,000 annually.

Institutional and Government Engagement

The third strategy of promotion is to engage institutions or government entities. Development finance institutions and governments have significantly stepped up their investment in modernizing agriculture. The African Development Bank has approved about \$11 billion for 2024 and 2025, focusing on farming and climate resilience. The World Bank has planned to increase funding for farming to a total of \$9 billion annually by 2030 (Reuters, 2024).

The agricultural drone service also contributes to climate-smart agriculture, which seeks to improve irrigation, reduce pesticide use, and optimize crop yields. An allocation of around \$3,750 annually will be set aside for engaging institutions, which includes policy

briefings, presentations at the ministry level, and participating in government-backed agricultural expos. Engaging institutions has the potential for accessing pilot projects or regional contracts with significantly reduced entry risks for the agricultural drone service.

Selecting an International Pricing Strategy (Justin)

A global pricing strategy for this agricultural drone service can reasonably follow a “market-based, value-oriented” approach with regional differentiation rather than a single uniform price. Industry sources suggest that agricultural drone spraying services are often priced on a per-hectare basis, with reported ranges that reflect local labor costs, input prices, and farm sizes; some African pilots note that drones can spray roughly 10 acres per hour and may enable per-acre service fees that remain competitive with manual or tractor-based spraying while still recovering equipment and operating costs (Rural Hack, 2025; IA Precision, 2025). At the same time, the global agriculture drones market is projected to grow at CAGRs above 20–30%, which indicates rising competition from both hardware vendors and local service providers; this environment favors a pricing strategy that emphasizes total cost of ownership and return on investment for farmers, rather than only matching rivals’ fees (Allied Market Research, 2025; MarketsandMarkets, 2025).

Given estimated start-up and operating costs (equipment, regulatory compliance, software, staff, and transport), a practical structure for foreign markets could combine tiered per-hectare pricing with volume discounts and bundled services. For example, the firm could set base rates that target a positive contribution margin after accounting for depreciation of drones and typical flight hours per season, then offer lower per-hectare prices for cooperatives that commit larger areas or multi-season contracts (GreenCape, 2025; IA Precision, 2025). Where donor programs or agribusinesses co-finance digital agriculture, the company may negotiate institutional contracts that partially subsidize smallholder fees, which aligns with documented arrangements in African agtech where development partners support

the adoption of precision tools (Market Data Forecast, 2025; IFPRI, 2025). Because economic conditions vary across countries—such as differences in inflation, fuel prices, and exchange rates—the company may also need to apply a “flexible regional pricing” element, adjusting nominal prices while maintaining a target margin band and communicating clearly how pricing reflects service scope and local cost structures (GreenCape, 2025; Allied Market Research, 2025). Overall, a value-based, regionally adapted pricing strategy that links per-hectare fees to demonstrable savings in inputs and labor, documented through pilots and yield data, appears consistent with reported farmer decision-making and current competitive dynamics in the agricultural drone service market (IA Precision, 2025; Rural Hack, 2025).

Phase 4 – Implementing the Global Market Plan

Determining Financial Results and Measuring International Business Success (Zach)

In determining the financial results of a firm, an important focus must be placed on the appropriate financial figures. For the proposed agricultural drone business, several financial metrics will provide insight into the financial performance and projectability of the business, illustrating key areas where the business will need to perform to continue operating. The first metric is the LTV:CAC ratio, which measures the lifetime value of a customer compared to the cost to acquire that customer. This metric retains significant importance to selling drones in African markets because of the nature of customer acquisition costs in foreign markets. The complications of limited infrastructure and low trust among local entities in new African markets drive up the cost of customer acquisition (Yibeltal, 2023). Consequently, if the CAC ratio is extreme, then the LTV ratio must increase in order to balance the ratio. The primary method by which this will be accomplished is through the onboarding of new cooperative smallholders, who will eventually be converted into whole product consumers. This revenue model supports a higher LTV because it serves consumers

at varying levels of trust and willingness to pay and seeks to move consumers up the value chain.

Next, the business should observe the contribution margin per hectare. Contribution margin is derived from subtracting the variable costs from total revenue, and it can serve as an important indicator in financial performance. For an agricultural drone business, the contribution margin per hectare will reveal how much the variable costs of traveling to remote farms, repairing damaged drones, upgrading batteries, etc., are eating into our profits. Because the variable costs are small compared to the high fixed costs of developing such a high-tech drone, it is important to monitor the contribution margin on a per-hectare basis (Singh et al., 2025). The business will be able to evaluate profitability per farm, enabling identification of increased costs, because the costs might be difficult to detect otherwise amidst many hectares of coverage.

Another metric that needs to be monitored is the utilization threshold. Drones are a depreciating asset, meaning they begin to lose value as soon as they are purchased. The service life of a drone is actually so short that it is considered a perishing asset, typically lasting only three to five years (AgroNews, 2025). However, most models are retired due to the release of new models rather than mechanical failure, and the African consumers might not place a high importance on possessing the newest model every three years. Nevertheless, because of the short service life, utilizing or selling a high percentage of the drones produced is necessary to achieve financial success. The drones will depreciate at a flat rate, and time that the drones are not being used is time they are losing value. The business will seek to service most of its drones to cooperatives or sell the products before they can lose value as extra inventory.

The last metric will be the conversion rate of "service subscribers" to "asset owners," measuring how many subscribers eventually purchase a drone. The initial business structure

suggested that incorporating cooperative deals to maximize demand met without lowering the price point was not necessarily the long-term vision. Using a calculator provided by the University of Missouri and general cost assumptions outlined in the study, one must farm roughly 980 acres to justify the cost of purchasing agricultural drones rather than obtaining the required quantity through a subscription service (Singh, 2025). Applying this logic to farming in Africa, it will take farms of roughly the same size (about 380 hectares) to justify purchasing drones. Therefore, many small farms may opt to subscribe rather than purchase because of their size. The agricultural drone business will prioritize selling drones to larger farms rather than small ones, and the smaller farms might need to experience growth or an influx of capital before they decide to become asset owners. Regardless, tracking this conversion rate provides the business with important revenue forecasts that fund future business operations.

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